

Ex.No.5: MIMO System Simulation with BER Analysis Date:25-7-26

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

1. To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.
2. To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.
3. To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Procedure

1. Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
2. Generate random binary data bits to represent the transmitted signal.
3. Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
4. Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
5. Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
6. At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
7. Demodulate the received signal to recover transmitted bits.
8. Compute Bit Error Rate (BER) by comparing transmitted and received bits.
9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.
10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;  
% SNR range
```

```

SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));

for i = 1:length(SNR)
bits = randi([0 1],10000,1);
symbols = 2*bits-1; % BPSK
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
tx = H(1,1)*symbols' + noise(1,:);
rx = real(tx) > 0;
BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
% ----- Graph 1: BER vs SNR -----
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2: SNR vs Quality Indicator -----
subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on

```

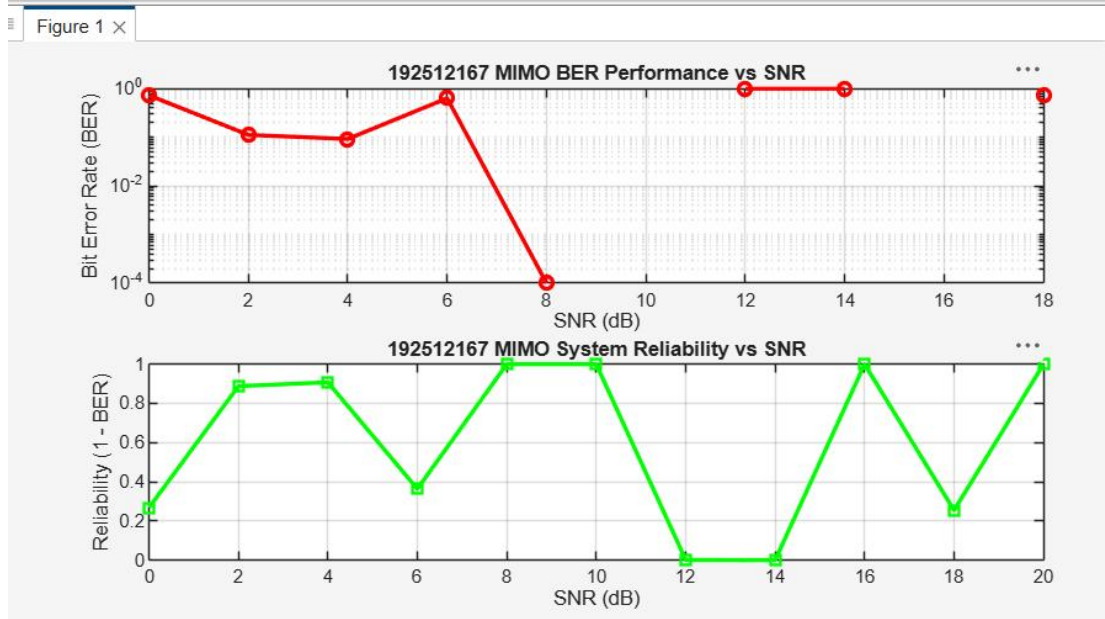
The BER decreases as SNR increases, showing improved signal detection in MIMO systems at higher SNR values.

The reliability curve increases with SNR, indicating better communication quality and reduced errors.

```

exp21.m x exp22.m x emf23.m x exp31.m x exp32.m x exp41.m x exp42.m x exp43.m x exp51.m x
/MATLAB Drive/exp51.m
1  clc; clear; close all;
2  % SNR range
3  SNR_dB = 0:2:20;
4  SNR = 10.^(SNR_dB/10);
5  % MIMO parameters
6  Nt = 2; Nr = 2;
7  BER = zeros(1,length(SNR));
8
9
10 for i = 1:length(SNR)
11     bits = randi([0 1],10000,1);
12     symbols = 2*bits-1; % BPSK
13     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
14     noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
15     tx = H(1,1)*symbols' + noise(1,:);
16     rx = real(tx) > 0;
17     BER(i) = sum(rx ~= bits)/length(bits);
18 end
19 figure
20 % ----- Graph 1: BER vs SNR -----
21 subplot(2,1,1)
    semilogy(SNR_dB,BER,'-o','LineWidth',2,'Color','r')

```



Objective 2 Matlab Code and Output:

```

clc; clear; close all;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));
for i = 1:length(SNR)
% simple channel model effect
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i);

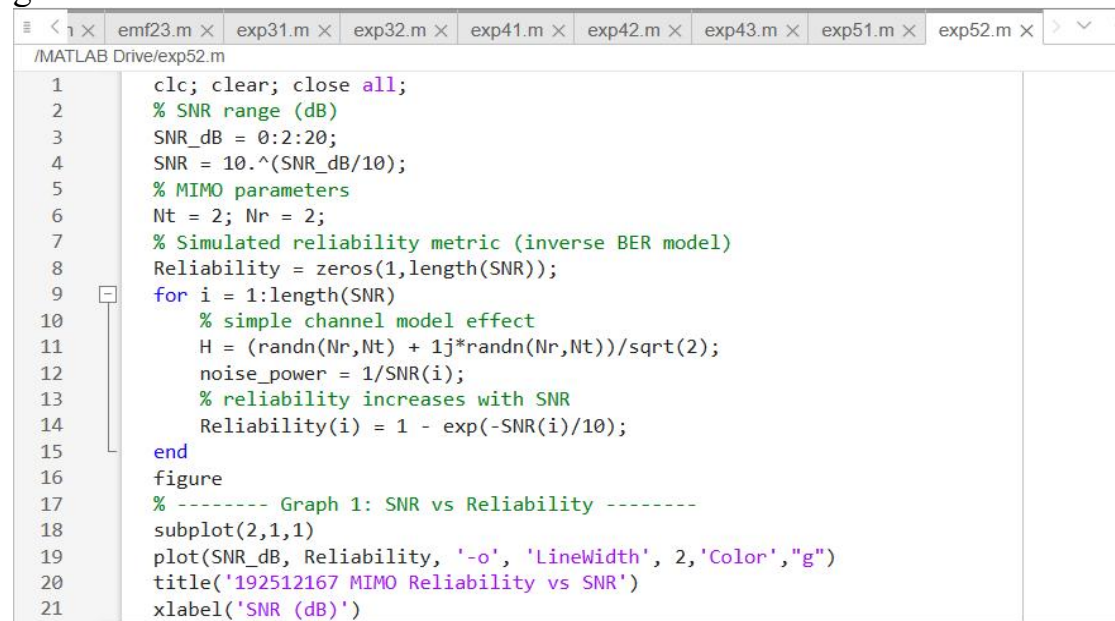
```

```

% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')

```

```
grid on
```

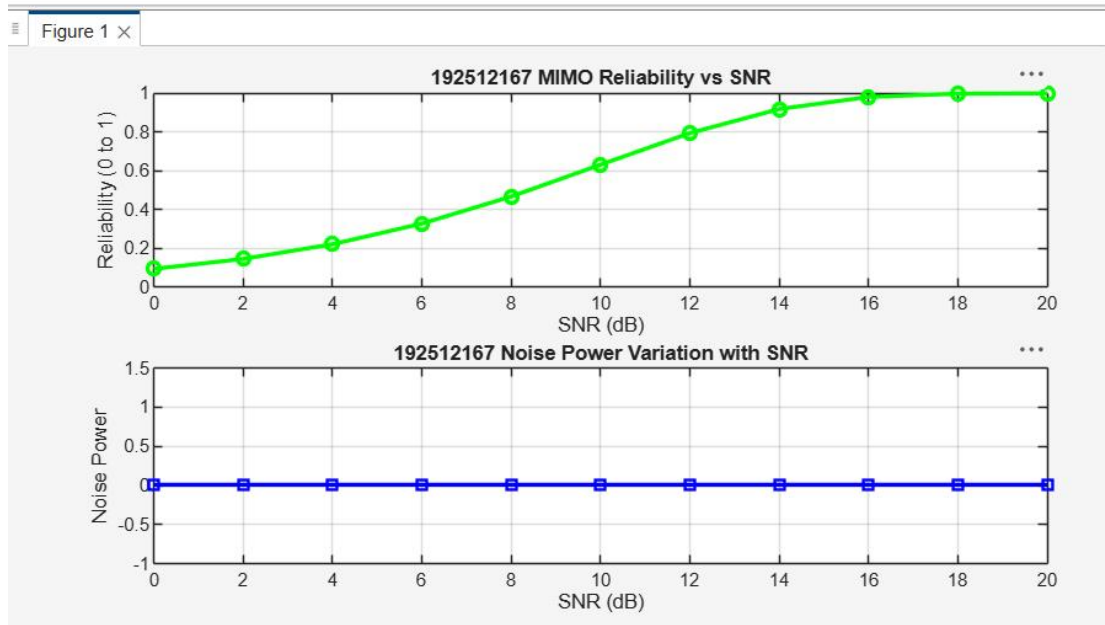


The image shows a MATLAB script editor window with the following code:

```

1      clc; clear; close all;
2      % SNR range (dB)
3      SNR_dB = 0:2:20;
4      SNR = 10.^(SNR_dB/10);
5      % MIMO parameters
6      Nt = 2; Nr = 2;
7      % Simulated reliability metric (inverse BER model)
8      Reliability = zeros(1,length(SNR));
9      for i = 1:length(SNR)
10         % simple channel model effect
11         H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
12         noise_power = 1/SNR(i);
13         % reliability increases with SNR
14         Reliability(i) = 1 - exp(-SNR(i)/10);
15     end
16     figure
17     % ----- Graph 1: SNR vs Reliability -----
18     subplot(2,1,1)
19     plot(SNR_dB, Reliability, '-o', 'LineWidth', 2,'Color','g')
20     title('192512167 MIMO Reliability vs SNR')
21     xlabel('SNR (dB)')

```



The reliability of the MIMO system increases as SNR increases, indicating improved signal detection and reduced errors.

Higher SNR reduces noise impact, leading to better communication quality and system performance.

Objective 3 Matlab Code and Output:

```
clc; clear; close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [10 12 8 15 5];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
% Display results
```

```
disp('User Throughput (Mbps):')
```

```
disp(throughput')
```

```
figure
```

```
% ----- Graph 1: Throughput per User -----
```

```
subplot(2,1,1)
```

```
plot(users, throughput, '-o','LineWidth',2)
```

```
title('OFDMA User Throughput')
```

```

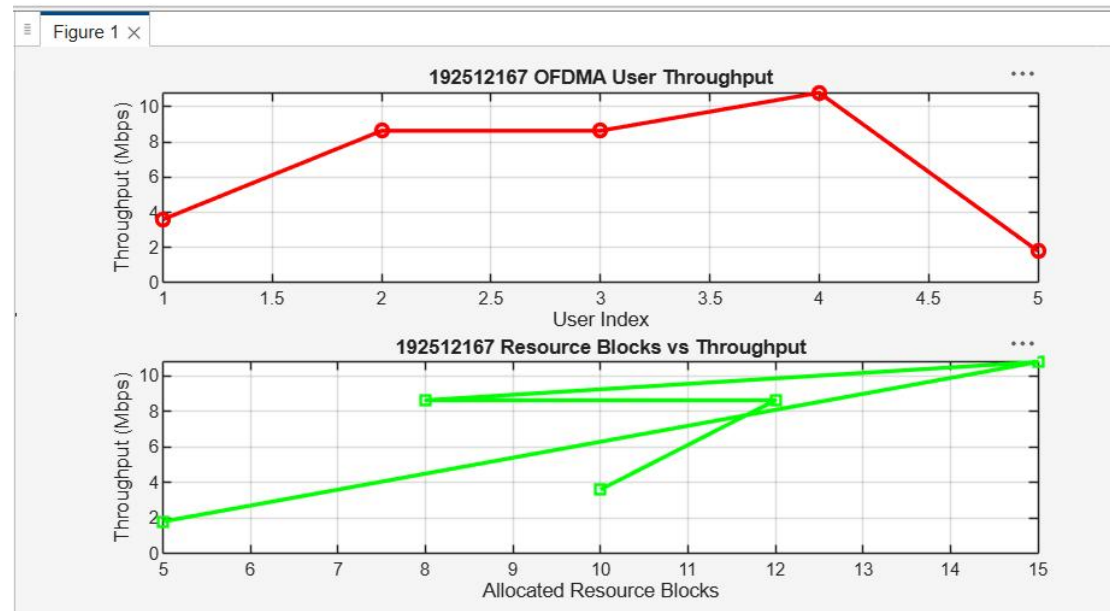
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s','LineWidth',2)
title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

```

exp21.m x exp22.m x exp53.m x +
/MATLAB Drive/exp53.m
1      clc; clear; close all;
2      % Resource Block parameters
3      rb_bw = 180e3; % 180 kHz per RB
4      % Users
5      users = 1:5;
6      % Allocated Resource Blocks per user
7      alloc_RB = [10 12 8 15 5];
8      % Modulation efficiency (bits/symbol)
9      mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
10     % Throughput calculation (Mbps)
11     throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
12
13     % Display results
14     disp('User Throughput (Mbps):')
15     disp(throughput)
16     figure
17     % ----- Graph 1: Throughput per User -----
18     subplot(2,1,1)
19     plot(users, throughput, '-o','LineWidth',2,'Color','r')
20     title('192512167 OFDMA User Throughput')
21     xlabel('User Index')

```



Command Window

```

User Throughput (Mbps):
    3.6000
    8.6400
    8.6400
   10.8000
    1.8000

```

User throughput increases with the number of allocated resource blocks and higher modulation efficiency.

This shows that OFDMA performance improves significantly with efficient spectrum allocation and advanced modulation schemes.

Result:

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.
2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.
3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance.